Independent audit of Phase 2i: normalisations, numerical evidence and unresolved identifications

Quantum Boussinesq solitons project

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Abstract

(c) Phase 2i contains useful new conditional asymptotic calculations and correct orientation bookkeeping, but its printed Wiener–Hopf theorem is incorrectly normalised, its exchange criterion is false, and its assertion that all six audits are resolved is unsupported. The original checker passes despite these problems. Independent inverse Laplace transforms support the stated momentum coefficients after a missing normalisation is restored. Independent phonon quadratures confirm the orientation formulas, not their interpretation as a physical determinant normalisation. **(d)** The physical quantum frame and the boundary-compatible projected determinant remain unestablished.

1 Scope and reproducibility

Labels retain their repository meanings: (a) established in literature, (b) literature heuristic, (c) project derivation or computational evidence, (d) unestablished interpretation or research judgment. All new mathematical checks below are in `scripts/audit_phase2i.py`; numerical evidence is explicitly distinguished from a proof. No existing file was edited, no git operation performed, and no regularisation, physical charge or phase convention selected. The good-equation conventions and the existing Toda extraction are retained. The independent quantum-frame investigation is outside this audit's scope.

(c) The Phase 2i source and PDF, the six named audits, the Phase 2f/2h definitions, and relevant checkers were read. The Phase 2h independent audit was read in full before implementation. This checker imports no Phase 2i formulas or audit checker. Only the expressly permitted `FastNodesEdgeBA` finite-h solver is reused; its endpoint is set to exactly q, and its root bracket is expanded when necessary. Analytical and integral checks use 60-digit mpmath; convolution quadrature uses 35 digits; the dense Bethe solver and data regression use float64. JSON stores raw data, input hashes, fit variants and timings. Hashes are checked again at the end. Finite-h results are cached; use `python3 audit_phase2i.py --refresh` to regenerate them.

The terms *confirmed*, *qualified*, and *refuted* refer to the specific claim, rather than the entire programme. In particular, "confirmed conditional algebra" does not certify a physical S-matrix. **(d)**

2 Theorem 1: three distinct normalisation problems

Fourier symbol: refuted as printed (`audit_phase2i.py`, `wh`). **(c)** Write $T(s) = \widetilde{\Phi}_{\mathrm{WH}}(s)$ with Phase 2i's factor. For $\omega \neq 0$, direct multiplication gives

$$\Phi_{\mathrm{WH}}(i\omega)\Phi_{\mathrm{WH}}(-i\omega) = \frac{\pi e^{|\omega|/2}}{\sinh(|\omega|/2)} = \frac{2\pi}{1 - e^{-|\omega|}}, \tag{1}$$

not $2\pi^2 e^{\pi|\omega|}/\sinh(\pi|\omega|)$. The digamma integral representation gives the same nonzero-frequency result: the cosine term contributes $2\pi/(e^{|\omega|} - 1)$ to the transform of $-2\,\mathrm{Re}\,\psi(1 + ix)$, in addition

to 2π from the delta. The transform is distributional at zero frequency; this calculation does not fix a zero-frequency subtraction or justify dropping the finite-sea constant $2\log h$. The large-frequency limit is 2π, whereas the printed symbol approaches $4\pi^2$. Four high-precision product checks reproduce the discrepancy. The original script never tests this symbol. Its subsequent symbolic series uses T, so this particular symbol misprint does not propagate into that series.

Momentum Laplace transform: refuted as printed. (c) Let $G(u) = \frac{4}{3}y(u)^{3/2}$. Since $y(u) \sim u$, its leading Laplace coefficient must be $\Gamma(5/2)\,4/3 = \sqrt{\pi}$. The normalisation compatible with the implemented expansion is

$$\mathcal{L}G(s) = \sqrt{\pi}\,T(s)/s^2, \tag{2}$$

not $T(s)/s^2$. Without this factor the leading result is $y(u) \sim \pi^{-1/3}u$, incompatible with the asserted $\delta u = O(\log y)$. The script silently multiplies every inverse-Laplace term by `sp.sqrt(sp.pi)` in `rhs_u`; this is not in the displayed theorem. Also, the printed scale $aP_{\rm rel}/(\sqrt{bh}\,h)$ is inconsistent with $P_{\rm rel} = 4b(\kappa\cosh\kappa - \sinh\kappa)/a$ and $\kappa^2 = yh/b$: the scale giving $4y^{3/2}/3$ is $a\sqrt{b}\,P_{\rm rel}/h^{3/2}$. These are algebraic normalisation tests, not alternative physical conventions.

The printed inverse series is also refuted. (c) Put $A = 1/(2\pi)$, $c = c_{\rm edge}$, $d = d_0$, $e = e_0$. Reverting $u = y - A\log y + c + d/y + e/y^2$ necessarily gives

$$y - u = v_0 + v_1/u + v_2/u^2 + \cdots, \tag{3}$$

$$v_0 = A\log u - c, \qquad v_1 = Av_0 - d,$$

$$v_2 = Av_1 - \tfrac{A}{2}v_0^2 + dv_0 - e. \tag{4}$$

In particular the $u^{-2}(\log u)^2$ coefficient is $-1/(16\pi^3)$. Phase 2i's equation `y_of_u` has no such term. Numerical inverse Laplace transforms independently contradict that displayed equation. The script computes an intermediate inverse series but never compares it to the printed `y_of_u`; its final reversion can therefore pass.

Coefficients: qualified as a theorem; strong conditional confirmation. (c) With (2), independent numerical inversion supports $c = (1 - \log(8\pi))/(2\pi)$, $d = 1/48 - 1/(4\pi^2)$ and $e = (4 + 6\zeta(3) - \pi^2)/(96\pi^3)$. Fitting the independent inverse transform without setting logarithmic coefficients to zero gives

$$(d_1, e_2, e_1) \simeq (-2.4\,10^{-21}, -2.5\,10^{-15}, 7.2\,10^{-14}), \qquad e_0 \simeq 0.000451097178279.$$

This fit uses sixteen u values from 1000 to $1000(1.5)^{15}$ and includes all logarithmic terms through u^{-4} in the fitting basis. It is a numerical check of the corrected transform, not an independent proof of exact zeros. The original formal SymPy reversion supports exact zeros *conditional* on its transform input. Independent checks at $u = 8, 32, 128, 512, 2048, 8192$ find the residual after e/y^2 smaller than $0.0004/y^3$.

(d) Matching a homogeneous edge solution to a fixed bulk density is a legitimate way to determine its otherwise free amplitude; it is not intrinsically an arbitrary convention. Here the projection, zero-frequency matching and normalisation are asserted rather than worked through, and two printed normalisations contradict that matching. Hence Phase 2i does not supply the claimed complete derivation from the finite-h equation. The independent finite-h test below is additional evidence, not a replacement for it.

3 Theorem 2: convolution, contact terms and asymptotics

The double Laplace formula is refuted as printed (`audit_phase2i.py`, `wh`, `finite`). (c) In the corrected convention set $B = G'$ and $f = G''$ on the open half-line. Stirling's limit gives $B(0^+) = \sqrt{\pi}$, and therefore

$$\mathcal{L}B = \sqrt{\pi}T/s, \qquad \mathcal{L}f = \sqrt{\pi}(T-1).$$

Differentiating the printed convolution $2\int_0^{u_2} f(u_1 - w)B(u_2 - w)\,dw$ at $u_1 > u_2$ retains the endpoint $2\sqrt{\pi}f(u_1 - u_2)$. Equivalently, use the causal distribution $\sqrt{\pi}\delta + f$, whose Laplace transform is $\sqrt{\pi}T$. The compatible full double transform is

$$2\pi\frac{T(s_1)T(s_2) - 1}{s_1 + s_2}, \tag{5}$$

with its contact term specified, rather than Phase 2i's prefactor 2. Thus the printed Laplace pair and the classical normalisation of its convolution cannot both hold. The original `verify_2` does not evaluate an inverse transform, the two-variable transform, or a finite-h dressed phase.

Convolution after correction: qualified, numerically supported. (c) We numerically invert (2) and integrate the first convolution directly, with 24- and 48-node Gauss quadrature. We compare with $[\Phi(q + hu, q + hv) - \Phi(q + hu, q)]/h$ from the finite-sea solver at $(u, v) = (4, 1), (8, 2), (8, 4)$. Both arguments are fixed before $h \to 0$; the corresponding (Y, r) are obtained from the independently computed $y(u)$. The finite-h grids, extrapolation spread and numerical results are reported in the accompanying tables and JSON. This is a test of the same two-term normalisation used in Phases 2f/2h, not an antisymmetrised substitute.

Leading integral and c_1: confirmed conditional evaluations (`audit_phase2i.py`, `integrals_data`). (c) Independent quadrature verifies

$$\frac{4}{\pi}\int_0^r \log(s/r)\frac{s^2}{1 - s^2}\,ds = -\frac{4}{\pi}(\chi_2(r) - r)$$

and the displayed c_1 integral against the stated elementary expression, at five ratios between 0.2 and 0.8, with errors below 10^{-54}. These are useful conditional integral identities. The balanced logarithmic coboundary cancels $\log Y$ and c at fixed $t > 0$ algebraically. **(d)** It does not alone prove a uniform integrated expansion: Yt^2 is not large near $t = 0$. The asserted $O(Y^{-2})$ remainder and absence of other endpoint terms require an endpoint estimate or matched expansion. Replacing v by $y(v)$ makes the exact measure regular but does not make a large-y expansion uniform at $y = 0$. Thus the leading limit and two-loop coefficient are qualified as conclusions about the Bethe equation, although their formal integrals and numerical evidence survive.

The claimed four-term reciprocity resolution is refuted as printed. (c) A four-term edge-subtracted phase vanishes when either argument is zero. The asserted symmetric product $2B(u_1)B(u_2)$ does not: $B(0^+) = \sqrt{\pi}$. This is a direct boundary-condition contradiction. The earlier Toda dictionary audit explicitly distinguishes two-term and four-term phases by one-body edge terms; Phase 2i's remark suppresses that distinction. Its script checks only a series of the proposed product in $1/Y$, not the endpoint identity or its finite-h limit. **(d)** No replacement physical quantum frame is selected here.

4 Saved Toda data and the finite-h error budget

(c) The two-term Phase 2f definition is

$$R_h = \{\Phi(p_1, p_2) - \Phi(p_1, e_2) - M_1(P_{n2}/a - \Pi_2)\}/h - 2Y(1 - r^2)\log\frac{1-r}{1+r}.$$

Phase 2h defines its R_0 as the $h \to 0$ limit of the same expression, equivalently $\Phi_q(u(Y), u(r^2Y)) - \Phi_{\rm cl}(Y, r^2Y)$. The Phase 2i saved-data subtraction retains this definition in the limit: the finite-h classical Toda phase has the same leading term. It is a numerical improvement to the extrapolation, not a new quantum frame. Our independent high-precision subtraction uses the hyperbolic cross ratio

$$A_T = \frac{\sinh^2((\kappa_1 - \kappa_2)/2)}{\sinh^2((\kappa_1 + \kappa_2)/2)},$$

and the classical momentum and topological term, avoiding the cancellation in the source script's Hirota-polynomial evaluation.

(c) The original script fixes c_0 to its proposed value and computes

$$\widehat{c}_1 = 64[R_0(32) - c_0] - 16[R_0(16) - c_0]. \tag{6}$$

This cancels a Y^{-2} term *if* the proposed c_0 and a power expansion are correct. It is different from the old two-point slope $32[R_0(16) - R_0(32)]$, which estimates c_0 freely but neglects curvature. Consequently its agreement is not an independent simultaneous determination of c_0 and c_1. The numerical tables compare both estimators and free quadratic $1/Y$ fits on $Y = 8, 16, 32$ and $4, 8, 16, 32$. No fit uses the target c_1 as input.

(d) Moving the edge to $q(h)$ does not prove analytic dependence on h. Subtraction of known classical curvature is justified; rejection of every $h \log h$ correction requires an asymptotic argument about the quantum finite-sea error. Fit spreads below are sensitivity diagnostics, not statistical confidence intervals or rigorous truncation bounds. The finite-h coefficient estimates do not resolve exact vanishing of logarithmic coefficients.

Fresh finite-h momentum check (c). The grids are $h = 0.04, 0.02, 0.01, 0.005$, with respectively $681, 1062, 1824, 3348$ Gauss nodes. A separate $h = 0.01$ solve uses 2189 nodes. The table compares the four-point cubic intercept to the finest-three-point quadratic intercept; mesh differences are measured at $h = 0.01$.

y	Cubic δu	Difference from asymptotic	Power-fit spread	Mesh difference
2	-0.4663923834	4.90e-05	2.52e-09	1.01e-11
4	-0.5757147346	4.79e-06	1.27e-08	1.39e-11
8	-0.6854957971	5.52e-07	6.62e-08	1.68e-11
16	-0.7955382763	9.91e-08	3.71e-07	1.89e-11
32	-0.9057166221	3.45e-07	2.19e-06	2.04e-11
64	-1.0159613570	3.48e-06	1.29e-05	2.15e-11

(c) At $y = 16$, extracting e_0 from this intercept gives 0.00047646, while the power-fit spread propagated by y^2 is about 0.000095. At $y = 32, 64$ that amplification is still worse. Thus finite-h data support c, d but do not independently resolve e_0 or exact logarithmic zeros. The corrected inverse transform provides the sharper conditional test.

Fresh convolution comparison (c).

(u, v)	(Y, r)	Convolution	BA cubic	Discrepancy	Power-fit spread
$(4, 1)$	$(4.598, 0.554)$	2.19960638	2.19960612	2.6e-07	5.3e-06
$(8, 2)$	$(8.699, 0.536)$	3.78568886	3.78568487	4.0e-06	4.6e-05
$(8, 4)$	$(8.699, 0.727)$	10.08223894	10.08222807	1.1e-05	1.3e-04

(c) Quadrature changes between 24 and 48 nodes are below $2.6\,10^{-16}$; these are float-node quadrature comparisons, not 35-digit accuracy claims. The BA discrepancies are smaller than the power-extrapolation spreads. The independent mesh comparison changes the three finite-h phases by at most $3.7\,10^{-11}$ at $h = 0.01$.

Saved-data sensitivity after classical subtraction (c). All following numbers use the same independently subtracted saved data. The fixed-c_0 estimator is Eq. (6). "Free two" is the unconstrained intercept/slope estimate at $Y = 16, 32$.

r	h extrapolation	Fixed-c_0 c_1	Free two c_1	Target
0.4	cubic4	-0.0078316	-0.0072586	-0.0077366
0.4	quad3	-0.0083604	-0.0070237	-0.0077366
0.4	quad4	-0.0102111	-0.0062013	-0.0077366
0.4	hlog3	+0.0065531	-0.0128715	-0.0077366
0.4	hlog4	+0.0233713	-0.0191761	-0.0077366
0.5	cubic4	-0.0215719	-0.0213581	-0.0214446
0.5	quad3	-0.0225280	-0.0209332	-0.0214446
0.5	quad4	-0.0258741	-0.0194462	-0.0214446
0.5	hlog3	+0.0051142	-0.0317775	-0.0214446
0.5	hlog4	+0.0365390	-0.0435828	-0.0214446
0.6	cubic4	-0.0486099	-0.0488987	-0.0484360
0.6	quad3	-0.0501045	-0.0482344	-0.0484360
0.6	quad4	-0.0553354	-0.0459094	-0.0484360
0.6	hlog3	-0.0055121	-0.0657401	-0.0484360
0.6	hlog4	+0.0456838	-0.0850237	-0.0484360

(c) "cubic4" uses $0.02, 0.01, 0.005, 0.0025$; "quad3" and "hlog3" use the finest three; "quad4" and "hlog4" use all four. The logarithmic basis is $(1, h, h\log h)$. Pure-polynomial alternatives alone shift the fixed-c_0 estimates by more than $2\,10^{-4}$; logarithmic alternatives still reverse signs. Hence the claim that subtraction resolves the earlier fit ambiguity is refuted, although the selected cubic comparison passes.

r	Cubic h fit, free quadratic $1/Y$ (3 points)	Same (4 points)	Fixed-c_0 spread
0.4	-0.0079815	-0.0078174	0.03358
0.5	-0.0213367	-0.0215472	0.06241
0.6	-0.0480635	-0.0477384	0.10102

The last column includes all six saved-data variants, also the coarser four-point cubic grid $0.04, 0.02, 0.01, 0.005$; it is a sensitivity range. No statistical or rigorous error interpretation is assigned. **(c)**

5 Theorem 3: correct bookkeeping, false necessity, no physical selection

The "iff" criterion is refuted (`audit_phase2i.py`, `kp`). (c) Inversion antisymmetry is sufficient for antisymmetry of a ratio-kernel bilinear under exchange; oddness is unnecessary for that argument. For example $K(z) = z^2 - z^{-2}$ is even and inversion-antisymmetric and gives antisymmetric functionals. More decisively, all soliton and defect weight vectors have zero total weight. Adding a constant to $\widetilde{K}_L$ changes neither $\mathcal{F}$ nor $\mathcal{I}_{SS}$, but destroys both claimed kernel identities. This refutes necessity even while retaining *both* functional identities universally. A criterion modulo the null space, with additional domain assumptions, would be a different statement.

Orientation identities are confirmed; physical interpretation qualified. **(c)** We reconstruct the elementary arctangent phonon phase independently, then integrate $\theta_1\theta_2'/(4\pi)$ on the two open half-lines. For $(\alpha_1, \alpha_2) = (0.3, 0.7)$ all four orientations satisfy

$$I_{12}^{\mathrm{raw}} = \mathcal{I}_{SS}[\widetilde{K}_L] - \epsilon_1\pi/2, \qquad J_{12} = \mathcal{I}_{SS}[\widetilde{K}_L] - (\epsilon_1 - \epsilon_2)\pi/4.$$

We separately integrate the rational derivative of the defect transmission phase for $q_D = 0.4, 0.1, -0.45$, both soliton directions and both defect orientations. These cover $s < -1$, $|s| < 1$, $s > 1$, and confirm

$$I_{SD}^{\mathrm{raw}} = \mathcal{I}_{SD}[\widetilde{K}_L] + \epsilon\Delta\theta_D/4.$$

The maximum discrepancy is below 10^{-58} (recorded fully in JSON), with tolerance 10^{-50}. Thus the offsets found in the old audit are explained exactly. They have been moved into explicit orientation constants, not removed from the raw integrals. **(d)** This is an algebraic redefinition; it does not calculate an absolute determinant phase, select I versus J, or normalize defect states. The real-part kernel used on complex arguments is not itself holomorphic. Holomorphic continuation uses the corresponding branchwise analytic expression; that distinction must be maintained.

6 Bootstrap synthesis and defect spectrum

Crossing confirmed; printed Darboux identity refuted (`audit_phase2i.py`, `kp`). **(c)** Independent root sums and the canonical/mode charge shifts check both angle substitutions. But the printed Darboux defect delay has an extra $\sqrt{3}$. With the repository roots $q_{2,a} - q_{2,b} = k_2$ and the independently reconstructed raw delay $\Delta x_D = -\epsilon_1 \log|(p_{1,b} - q)/(p_{1,a} - q)|/k_1$, the right side of the printed coboundary exceeds its left side by

$$2k_2\gamma_1(\sqrt{3} - 1), \qquad \gamma_1 = -\frac{3\cos 2\alpha_1}{\sin\alpha_1 \sin 3\alpha_1}. \tag{7}$$

Replacing its shift $-2\sqrt{3}\epsilon_1\eta\gamma_1 q$ by $-2\epsilon_1\eta\gamma_1 q$ gives the claimed identity in all four tested orientations. This is an algebraic correction within the existing root normalisation, not a new physical convention. The original checker never constructs a defect delay. **(d)** Crossing and this corrected identity do not determine a physical fusion constant or supply stable intermediate states.

Breather continuation: qualified. **(c)** Direct continuation of the same three soliton roots used on the real axis gives $\mathrm{Re}\,\mathcal{I}_{SS} = 0$ and $\mathrm{Im}(\Delta P_{\mathrm{can},1} + \Delta P_{\mathrm{can},2}) = 0$ at both quoted conjugate pairs, to 10^{-50}. Branch-point cancellation at $z = 1$ is independently checked. The original script instead uses angles $\alpha \pm \pi/3, \alpha$ in this subsection, different from its real-soliton $\alpha \pm 2\pi/3, \alpha$ roots. The vanishing real part can survive this change because conjugation and exchange symmetry enforce it; that test alone does not test the physical root continuation. **(d)** Pure imaginary phase implies a positive multiplier only where that phase is finite. The sampled identities are not a global proof of pole freedom or a quantum quantisation rule. The shifted Bogdan–Charkina condition is asserted in the text and is absent from the checker; neither a corrected breather action nor its charge/frame identification is derived. It remains (d).

Defect correction: confirmed, with a threshold qualification (`audit_phase2i.py`, `determinant`). **(c)** Independent differentiation of the nonlinear sewing conditions produces the linear matrix. Six off-shell determinants confirm the $-2/3$ prefactor. Filtering the four polynomial roots by $S > 1$ confirms the absolute-value formula for the second growth rate on either side of $s = \sqrt{3}$. At the switching point that rate is zero: there are not two strictly positive growth rates there. The corrected determinant and branch switch were already established in `audit_frames_defects`; they are successful adoption of prior (c) results, not new Phase 2i discoveries.

7 What happened to each finding of the six audits?

(c) The following is a textual coverage audit, not a revalidation of every formula in those six documents. "Relabelled only" means an outstanding task is acknowledged but not solved. Grouped entries combine findings with the same disposition; the explicit omissions are listed separately.

Prior finding	Phase 2i disposition
Independent Phase 2h audit	
Orientation-dependent π discrepancies	Resolved algebraically by explicit offsets.
Raw exchange sum and nonzero diagonal	Partially resolved: new kernel antisymmetrises a redefined phase.
Oddness alone does not imply antisymmetry	Partially resolved: inversion suffices; new "iff" is false.
Saved c_1 signs change with fit basis	Partially resolved: subtraction helps; robustness is not proved.
Leading limit previously built into the loop target	Partially resolved: conditional integral is new; WH derivation incomplete.
Classical $(\log Y)^2/Y$ Taylor term	Partially resolved: fixed-t cancellation; endpoint control absent.
Separated head-on asymptotics	Resolved in text; original Phase 2i script does not test them.
Wrong soliton profile	Resolved in text; original Phase 2i script does not test it.
Legendre identity versus physical charge	Relabelled only: field identification remains conditional.
Charges and action audit	
Regulated co-moving action needs cutoff endpoint	Not addressed by a new regulated calculation.
Canonical derivative requires on-shell profile and boundary variation	Relabelled only under general field-theory caveats.
Weighted cosine sum is not zero	Not addressed.
Printed sign of intermediate $dL_1/d\alpha$ formula	Not addressed.
Wrong profile	Resolved by the corrected profile.
Static defect has a localized quadratic momentum	Not addressed explicitly; no operator calculation.
Fifth-degree boundary potential; bulk tadpole insufficient	Partially resolved: potential acknowledged, counterterm open.
Conditional absorption-locus shift, not a charge switch	Not addressed; no quantum sewing/absorption calculation.
Determinant foundations audit	
Fourth-order domain/completeness differs from cited scalar argument	Relabelled only: full determinant remains open.
Wrong co-moving first-derivative sign	Not addressed.
Positive potential-field density check	Not addressed by a new check.
Orientation-dependent endpoints and exchange constants	Resolved algebraically, physical normalization unresolved.
Defect logarithm branch and normalized fusion states	Relabelled only.
Amplitude modes have tails; mixed symplectic term does not vanish	Partially resolved: tail term acknowledged; printed Darboux coefficient is wrong.
Endpoint Hessian is not field spectral flow or Maslov phase	Not addressed specifically; no spectral-flow calculation.
Unit Darboux Jacobian does not fix full functional measure	Relabelled only.

Prior finding	Phase 2i disposition
Equal localized endpoint frequencies do not prove cancellation	Not addressed; no localized-mode propagation.
Stable scattering domain and threshold completeness	Relabelled only; no spectral theorem.
Frames and defects audit	
Sewing determinant missing factor sixteen	Resolved.
Growing branch switch and $s = 2$ coalescence	Resolved, except marginal point wording.
KP triangle traverses unstable vacua	Relabelled only: physical fusion still open.
Edge reference cannot change a ratio slope	Partially resolved: acknowledged; conditional convolution supplies alternative origin.
Classical frame does not establish quantum frame	Not resolved; a proposed endpoint product is not the missing dictionary.
Toda dictionary audit	
Exact edge momentum and dressed-coordinate Jacobian	Not independently derived in Phase 2i.
Numerical $q(1 + 10^{-10})$ offset is nonuniform in h	Not addressed by its solver call.
Finite-sea reciprocity has endpoint terms	Partially resolved: invoked, but proposed four-term edge product fails boundary test.
Two-term versus four-term normalization matters	Not resolved; the remark conflates their symmetric remainders.
Classical frame leaves nonseparable symmetric remainder	Partially resolved: proposes an asymptotic expression without verifying identity.
Symmetric part does not fix antisymmetric physical frame	Not addressed by a Ward identity; still open.
Projected measure audit	
Tail-independent Wronskian determinant and collective volume	Not addressed explicitly; already proved in that audit.
Exact collision-time quadrature and conserved boundary flux	Not addressed.
Basis-independent projector	Not addressed.
Projector violates fixed potential endpoints; fixed-M rank loss	Not addressed beyond generic "measure open".
Solution tangents need not be temporal determinant zero modes	Not addressed.
Volume/endpoint map does not determine propagation phase	Not addressed.
Signed momentum derivatives need orientation factor	Not addressed explicitly.
Boundary-compatible projected determinant and absolute phase	Relabelled only.

8 Script audit, summary and failures recorded

(c) Running the original script from `phase2/scripts/` prints `ALL CHECKS PASSED` in 101.88 seconds. The symbol, printed inverse series, transform normalisation, double transform, convolution versus Bethe data, Darboux delay coboundary, shifted breather condition, and all-six-audit coverage are not tested as claimed. Its WH check inserts the missing $\sqrt{\pi}$; its convolution check evaluates formal coefficient integrals without constructing that convolution. Its endpoint check expands an assumed product. Its breather roots differ from the real-root implementation. The numerical momentum table cites four h values and $y = 32$, while the script actually uses three h values and stops at $y = 16$.

(c) The saved-data test is not literally a formula compared with itself: it uses independent

saved solver output. However it fixes c_0 to the target and adopts one extrapolation basis. Its $2\,10^{-4}$ discrepancy is a central-fit comparison, not a demonstrated total error bar. No evidence was found of historical tolerance loosening in the available single script snapshot; the present script cannot establish its editing history. The original failures list's claim that all prior failures were reproduced is false by inspection of the tests and the coverage table.

Claim	Verdict	Evidence and scope
Printed WH symbol, momentum transform and inverse series	Refuted (c)	Product, leading coefficient, reversion; `audit_phase2i.py wh`.
c, d, e and absent logarithmic coefficients	Qualified	Strong corrected-transform evidence (c); derivation and exact edge errors incomplete (d).
Printed double transform	Refuted (c)	Missing π after restoring momentum normalisation.
Correctly normalised convolution	Qualified	Independent inverse transform and finite-h comparison (c).
Leading residual and elementary c_1 integrals	Confirmed (c)	High-precision quadrature of formal integrals.
Exact Bethe asymptotics and strict remainder	Qualified	Missing uniform endpoint control (d).
Saved-data agreement below $2\,10^{-4}$	Qualified	Reproduced for one constrained estimator (c); fit sensitivity exceeds that precision.
Universal kernel "iff"	Refuted (c)	Constant null kernel and even inversion-odd counterexample.
SS/SD orientation formulas	Confirmed (c)	Independent raw quadrature, all requested orientations/regimes.
Kernel replacement derives physical phase	Qualified	Redefinition with explicit constants; physical normalization open (d).
Crossing / printed Darboux coboundary	Confirmed / refuted (c)	Crossing survives; delay shift has extra $\sqrt{3}$.
Breather reality and shifted quantisation	Qualified	Reality checked locally (c); physical shift asserted (d).
Defect determinant and branch correction	Confirmed (c)	Independent nonlinear sewing differentiation; marginal-point caveat.
Four-term symmetric endpoint product	Refuted (c)	Violates exact edge-subtraction boundary condition.
All six audits resolved	Refuted (c)	Finding-by-finding coverage table.

Failures recorded.

1. Prior-claim failures: wrong Fourier symbol, missing momentum-transform normalisation and physical scale, incompatible printed inverse series, double transform normalisation, false necessity criterion, extra $\sqrt{3}$ in the Darboux delay, and four-term endpoint product; unsupported blanket audit-resolution statement. Checks assert the existence of these discrepancies rather than asserting the false equalities.

2. The first independent solver run exceeded its inherited fixed momentum bracket at the largest coarse-h sample. The audit now doubles the bracket until it encloses the same root; no tolerance or equation was changed.

3. The first direct Darboux assertion failed. Its initial independent transcription also inverted the raw transmission ratio; this was corrected against the stated raw delay. The remaining discrepancy is exactly the printed extra $\sqrt{3}$, which the final check explicitly asserts. Neither the target equality nor a tolerance was weakened to hide the failure.

4. One report-assembly command used a repository-relative path while already inside `phase2/`, causing `FileNotFoundError`; rerunning from the repository root fixed the path without changing any result.

Measured execution and deliverables. (c) The final successful resumed checker ran in 106.01 seconds, reusing the completed finite-h and saved-data sections from the first run. Fresh dense Bethe solves took 352.46 seconds in total; the original Phase 2i checker took 101.88 seconds. The finite-h cache contains all five runs, with raw phase, momentum, node and normalization data. Independent convolution quadrature and data regressions completed before the initially failed Darboux assertion; the resumed run rechecks the WH diagnostics and all corrected phonon, crossing, breather, Darboux and sewing assertions. Its final line is `ALL CHECKS PASSED`. A normal run of `audit_phase2i.py` reruns every section, reusing only the dense-solver cache unless `--refresh` is specified. No existing implementation was changed; the full repository suite was not rerun. The new PDF was built twice with `pdflatex` and all pages were rendered and inspected. The TeX file includes its numerical tables and needs no separate table input.

References and bibliography leads. (c) No new external reference is cited or claimed read. This audit uses local project texts and independently checked mathematics. The bibliographic agent's requested leads remain: the Bogdan–Charkina title/equation designation differs between Phase 2i and the literature map; the Corrigan–Zambon article number differs between Phase 2i and the master report. These are unverified bibliographic leads here, not adjudicated reference corrections. No inference is drawn from the absence of the git-ignored literature folder.

What is new, and what remains. (c) Phase 2i contributes useful conditional higher-order WH reversion coefficients, an elementary formal $c_1(r)$ integral, improved classical lattice subtraction, and explicit all-orientation offset identities. It adopts the earlier determinant and spectrum corrections. **(d)** Its most important unresolved problem is a controlled, correctly normalised derivation linking the finite-sea Bethe phase to a physically defined quantum frame, including endpoint/contact terms and uniform edge asymptotics. Algebraic kernel choices cannot substitute for that identification or for the projected field determinant.